# Analog Electronics - Regan Dangol

## Assignment - 1.1

1. $V_1 = ?$, $V_2 = ?$

$V_1 = 4i$, $V_i = -2i$

Then,

KVL clockwise from bottom left

$$-32 + V_i - (-8) - V_2 = 0$$

$$-32 + 4i + 8 - (-2i) = 0$$

$$6i = 24$$

$$i = 4A$$

$$V_1 = 4i = 4 \times 4 = 16V$$

$$V_2 = -2(4) = -8V \quad //$$

2. Find $V_x$ and $V_o$

$V_x = 10i$

$V_o = -5i$

KVL clockwise

$-70 + V_x + 2V_x - V_o = 0$

$-70 + 10i + 2(10i) - (-5i) = 0$

$-70 + 10i + 20i + 5i = 0$

$35i = 70$

$i = 2A$

$V_x = 10(2) = 20V$

$V_o = -5(2) = -10V$

3, Equivalent Resistance (Req)

Right branch: $R_{S1} = 8+10+6 = 24\,\Omega$

First parallel block = $24\,\Omega$ is parallel to $12\,\Omega$

$$R_{P1} = \frac{24 \times 12}{24+12} = \frac{288}{36} = 8\,\Omega$$

Middle series block: $8\,\Omega$ is in series with the top $4\,\Omega$ resistor

$$R_{S2} = 8+4 = 12\,\Omega$$

Second parallel block: The $12\,\Omega$ is parallel with $6\,\Omega$ vertical resistor

$$R_{P2} = \frac{12 \times 6}{12+6} = \frac{72}{18} = 4\,\Omega$$

Final series block = $4\,\Omega$ is in series with the input $4\,\Omega$ and $3\,\Omega$ Resistor

$$Req = 4+4+3$$
$$= 11\,\Omega$$

# 4. Equivalent Capacitance ($C_{eq}$)

- Left parallel block : The 50 μF and 20 μF capacitor are in parallel.

$$C_{p1} = 50 + 70 = 120\ \mu F$$

- Right series block = The 60 μF and 120 μF are series

$$C_{s1} : \frac{60 \times 120}{60 + 120} = \frac{7200}{180} = 40\ \mu F$$

- Middle parallel block = 40 μF is parallel with the 20 μF vertical capacitor

$$C_{p2} = 40 + 20 = 60\ \mu F$$

- Final series block = The first equivalent (120 μF) and the new equivalent (60 μF) are in series

$$C_{eq} = \frac{120 \times 60}{120 + 60} = \frac{7200}{180} = 40\ \mu F$$

Q.5. Find the Voltage Across Each Capacitor.

1. Find $C_{eq}$: $60\,\mu F$ & $30\,\mu F$ are in series: $\frac{60 \times 30}{60+30}$

$= 20\,\mu F$

- This is in parallel with the $20\,\mu F$ capacitor $(V_2)$: $20+20 = 40\,\mu F$

- This is in series with the $40\,\mu F$ capacitor $(V_1)$

$$= \frac{40 \times 40}{40+40} = 20\,\mu F$$

2. Find the total Charge: $Q = C_{eq} \times V = 20 \times 150$
$= 3000\,\mu C$

3. Calculate Voltage:

$V_1 = Q/C_1 = 3000/40 = 75V$

- The remaining voltage drops across the parallel block: $V_2 = 150 - 75 = 75V$

- The charge on the rightmost branch $Q_{right}$
$= C_{series} \times V_2 = 20 \times 75 = 1500\,\mu C$

- $V_3 = 1500/60 = 25V$

$V_4 = 1500/30 = 50V$

The Ans - $V_1 = 75V$, $V_2 = 75V$, $V_3 = 25V$
& $V_4 = 50V$

Q-6. Find the Equivalent Inductance ($L_{eq}$)

1. The far-right 20H, 12H and 10H inductors are in series : $20+12+10=42$ H
2. This 42H is in parallel with the 7H inductor : $\frac{(42\times 7)}{42+7}=6$H
3. This 6H is in series with the top-left 4H and bottom-left 8H : $4+6+8=16$H

$$L_{eq}=18\text{H}$$

Q.7 : Calculate ladder Network Equivalent inductance

1. Far right : The top 40 mH and vertical 20 mH are in series : 40+20 = 60 mH

2. This 60 mH is in parallel with the 30 mH vertical inductor

$$\frac{60 \times 30}{60+30} = 20\,mH$$

3. Move left : This 20 mH is in series with the top 100 mH

$$100 + 20 = 120\,mH$$

4. This 120 mH is in parallel with the 40 mH vertical inductor

$$\frac{120 \times 30}{120+30} = 30\,mH$$

5. The left 30 mH is in series with the top 20 mH = 20+30 = 50 mH

6. Finally, this 50 mH is in parallel with the first vertical 50 mH inductor at the terminals $= \frac{50 \times 50}{50+50}$

$$= 25\,mH$$

$$L_{eq} = 25\,mH.$$